

given area—an inch in this case. For practical purposes, pixels per inch and dots per inch can be considered to be the same.

Going back to the analogy of the children with hats, if we were to say five children stand in one square metre (1m length x 1m height), the *children per metre* would be 5. If we know this number, and the total number of children, we can easily compute the area on the field that is occupied by the children. Similarly, knowing the dpi and the resolution, we can calculate the print size (that is, the actual physical size) of the image:

$\text{Resolution} / \text{dpi} = \text{size in inches.}$

Our screen is usually set at about 72 dpi, but good-quality prints need about 300 dpi. Photoshop saves at 72 dpi by default, but you can specify your own values. For high-quality prints, images should be at least at 200 dpi.

Photoshop allows you to change the resolution, and thereby the image size (if the dpi is kept same); it also allows you to change the dpi (and correspondingly the resolution, without changing the physical size)—or simply increase the size of the image. You can increase or decrease the size of the image by specifying

either the resolution or the actual physical size. Photoshop will do the corresponding calculation for the other values. You can scale up or down by specifying values or percentages of original. These options are all available in a single window called *Image Size*, under the Image menu.



The Image Size options in Photoshop